ECE 562: Deep Reinforcement Learning for Robotics

Fall 2026

Assignment 1: MDP Basics

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This assignment contains six questions organized into two sections:

- **Value Functions (40 points):** Definitions of state-value and state-action value functions, their mathematical relationships, and the implications of converting between them.
- **Bellman Equations (60 points):** Bellman expectation equations, Bellman operators, contraction properties, and exact policy iteration.

To earn full credit, clearly define all notation and assumptions, show the important intermediate steps in your derivations, and justify each conclusion. An answer consisting only of a final equation or conclusion, without adequate justification, may not receive full credit.

Unless otherwise stated, use the following assumptions:

- The state space $\mathcal{S}$ and action space $\mathcal{A}$ are finite.
- Rewards are bounded, and the discount factor satisfies $0 \leq \gamma < 1$.
- Policies may be stochastic unless a deterministic policy is explicitly specified.
- All inequalities between value functions are interpreted componentwise unless a norm is explicitly used.

Equivalent answers using different but mathematically consistent reward indexing or transition notation are acceptable.

Submission Instructions You may prepare your assignment using either of the following methods:

- Write your solutions and equations by hand. Photograph or scan all pages, ensure that the writing is clear and legible, and combine the pages into a single PDF file.
- Prepare your solutions in LaTeX. You may use the LaTeX template provided by the course or any other LaTeX template that you prefer and are comfortable using.

Submit your completed assignment through the course Canvas website. You may submit an unlimited number of times before the deadline. If you submit more than once, only your most recent submission will be graded. After submitting, check the uploaded file to ensure that all pages are included and readable.

1 Value Functions (40 points)

This section examines the fundamental value functions used in Markov decision processes (MDPs).

Q1 **(10 points)** Provide the formal mathematical definitions of the state-value function $V^\pi(s)$ and the state-action value function $Q^\pi(s, a)$ under a policy π.

Q2 **(10 points)** Derive the relationship between $V^\pi(s)$ and $Q^\pi(s, a)$.

Describe how to compute $Q^\pi(s, a)$ if the state-value function $V^\pi(s')$ is known for all possible subsequent states s'. Conversely, describe how to compute $V^\pi(s)$ if the state-action value function $Q^\pi(s, a)$ is known for all actions a.

Q3 **(20 points)** Based on the relationships derived in Question 2 of this section, analyze the conversion between V^π and Q^π in a model-free setting, where the environment's reward function and transition probabilities are unavailable.

(a) **(8 points)** Identify which of the following conversions is generally more difficult to perform exactly:

$$V^\pi(s) \longrightarrow Q^\pi(s, a) \qquad \text{or} \qquad Q^\pi(s, a) \longrightarrow V^\pi(s).$$

Explain why. Your explanation must identify the information required for each conversion. You may refer to the equations derived in Question 2 without deriving them again.

(b) **(12 points)** To illustrate your conclusion in part (a), construct a numerical counterexample showing that $V^\pi(s)$ and $\pi(\cdot \mid s)$ do not uniquely determine the individual action values.

Suppose that a state s has two actions and

$$\pi(a_1 \mid s) = \pi(a_2 \mid s) = 0.5, \qquad V^\pi(s) = 5.$$

Find two different pairs

$$\big(Q^\pi(s, a_1), Q^\pi(s, a_2)\big)$$

that are both consistent with the given state value. Choose the pairs such that

$$Q^\pi(s, a_1) > Q^\pi(s, a_2)$$

for the first pair, but

$$Q^\pi(s, a_1) < Q^\pi(s, a_2)$$

for the second pair.

Verify that both pairs produce $V^\pi(s) = 5$, and briefly explain why this example shows that $V^\pi(s)$ alone does not determine which action has the greater value.

2 Bellman Equations (60 points)

This section examines Bellman expectation equations, Bellman operators, their contraction properties, and exact policy iteration.

Q1 **(20 points)** Formally state the following Bellman expectation equations under a policy π:

(a) **(10 points)** The Bellman expectation equation for the state-value function $V^\pi(s)$.

(b) **(10 points)** The Bellman expectation equation for the state-action value function $Q^\pi(s, a)$.

Q2 **(20 points) Contraction of the Bellman Optimality Operator**

Consider a finite MDP with bounded rewards and discount factor $0 \leq \gamma < 1$. Let $R(s, a)$ denote the expected immediate reward and $P(s' \mid s, a)$ the transition probability.

The Bellman optimality operator T^* maps an action-value function Q to a new action-value function T^*Q. For each state-action pair (s, a), it is defined by

$$(T^*Q)(s, a) = R(s, a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s, a) \max_{a' \in \mathcal{A}} Q(s', a').$$

The supremum norm measures the largest absolute value of a function:

$$\|Q\|_\infty = \max_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} |Q(s, a)|.$$

Accordingly, $\|Q_1 - Q_2\|_\infty$ measures the largest absolute difference between two action-value functions.

Prove that T^* is a γ-contraction under this norm:

$$\|T^*Q_1 - T^*Q_2\|_\infty \leq \gamma \|Q_1 - Q_2\|_\infty.$$

Show the intermediate inequalities in your proof.

Hint: Begin by bounding $|(T^*Q_1)(s, a) - (T^*Q_2)(s, a)|$ for an arbitrary state-action pair. You may use

$$\left| \max_a x_a - \max_a y_a \right| \leq \max_a |x_a - y_a|.$$

Q3 **(20 points) Exact Policy Iteration**

Under the same assumptions as Question 2, use the contraction result you just proved to show why exact policy iteration approaches the optimal action-value function. Read the following background material and answer question (a) and (b).

(1) The Bellman expectation operator. A *Bellman backup* updates a value estimate using the expected immediate reward and the estimated value of what comes next.

For a fixed policy π, the Bellman expectation operator T^π maps an action-value function Q to a new function $T^\pi Q$. For each state-action pair (s, a),

$$(T^\pi Q)(s, a) = R(s, a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s, a) \sum_{a' \in \mathcal{A}} \pi(a' \mid s') Q(s', a').$$

To understand this calculation, start in state s and take action a. The environment produces an immediate reward and a next state s'. The next action a' is then selected according to $\pi(\cdot \mid s')$. The input $Q(s', a')$ estimates the return from that point onward.

The two sums average this estimate over the possible next states and actions. The result is discounted and added to the expected immediate reward. Thus,

$$\text{updated value} = \text{expected immediate reward} + \gamma \times \text{expected estimated remaining return}.$$

The word **"backup"** refers to using next-state action values to update the current state-action value. If Q_{old} is the current estimate, one backup produces

$$Q_{\text{new}} = T^{\pi} Q_{\text{old}}.$$

If the input Q is inaccurate, the result may still be inaccurate: one backup does not generally give the exact value of following π.

However, if the input is already the exact function Q^{π}, this calculation reproduces $Q^{\pi}(s, a)$ for every (s, a):

$$T^{\pi} Q^{\pi} = Q^{\pi}.$$

This is why Q^{π} is called a *fixed point* of T^{π}: applying the operator leaves it unchanged.

(2) Greedy policy improvement. The above mentioned Bellman operator does not improve the policy. How to improve the policy? We start with greedy policy improvement.

Exact policy iteration alternates between evaluating the current policy π_k exactly and choosing a new policy π_{k+1} that is greedy with respect to Q^{π_k}.

At every next state s', the greedy policy selects a maximizing action, or distributes probability only among maximizing actions. Therefore,

$$\sum_{a'} \pi_{k+1}(a' \mid s') Q^{\pi_k}(s', a') = \max_{a'} Q^{\pi_k}(s', a').$$

Substituting this equality into the definition of $T^{\pi_{k+1}}$, which is by definition, the greedy policy derived by choosing the action what results in the maximum Q^{π_k} value. We have

$$T^{\pi_{k+1}} Q^{\pi_k} = T^{*} Q^{\pi_k}.$$

Thus, a backup under the new greedy policy agrees with an optimality backup when both are applied to Q^{π_k}.

(3) Useful inequalities and fixed points equations. Let Q^{*} denote the optimal action-value function. You may use the following properties without proof:

- **Optimality:**

$$Q^{*} \geq Q^{\pi} \qquad \text{for every policy } \pi.$$

- **Optimal fixed point:**

$$T^{*} Q^{*} = Q^{*}.$$

- **Policy-improvement bound:**

$$T^{\pi_{k+1}} Q^{\pi_k} \leq Q^{\pi_{k+1}}.$$

  To see why, let $T = T^{\pi_{k+1}}$. The definition of greedy policy improvement $T^{\pi_{k+1}}$ and the old policy's fixed-point equation give

$$Q^{\pi_k} = T^{\pi_k} Q^{\pi_k} \leq T Q^{\pi_k}.$$

  Since T is monotone, repeatedly applying it preserves this ordering:

$$Q^{\pi_k} \leq T Q^{\pi_k} \leq T^2 Q^{\pi_k} \leq \cdots.$$

Here, T^n denotes n successive applications of T. Because T is a contraction with fixed point $Q^{\pi_{k+1}}$,

$$T^n Q^{\pi_k} \longrightarrow Q^{\pi_{k+1}}.$$

Taking the limit in $TQ^{\pi_k} \leq T^n Q^{\pi_k} \longrightarrow Q^{\pi_{k+1}}$. This compares one backup under the new policy with its fully evaluated action-value function.

All inequalities between functions are interpreted componentwise. For example, $Q_1 \leq Q_2$ means $Q_1(s,a) \leq Q_2(s,a)$ for every state-action pair (s,a).

Your tasks.

(a) **(16 points)** Prove

$$\|Q^* - Q^{\pi_{k+1}}\|_\infty \leq \gamma \|Q^* - Q^{\pi_k}\|_\infty$$

by completing the following steps.

i. **(4 points)** Use optimality and the policy-improvement bound to show

$$0 \leq Q^* - Q^{\pi_{k+1}} \leq Q^* - T^{\pi_{k+1}} Q^{\pi_k}.$$

Explain the direction of each inequality.

ii. **(4 points)** Use the greedy-policy equation and the optimal fixed-point equation to rewrite the upper bound as

$$Q^* - T^{\pi_{k+1}} Q^{\pi_k} = T^* Q^* - T^* Q^{\pi_k}.$$

iii. **(8 points)** Combine the preceding results, take the supremum norm, and finally, apply the contraction result from Question 2 to complete the proof of .

$$\|Q^* - Q^{\pi_{k+1}}\|_\infty \leq \gamma \|Q^* - Q^{\pi_k}\|_\infty$$

Then, explain why the componentwise nonnegativity established in step i permits the norm comparison.

(b) **(4 points)** Define the error after iteration k as

$$e_k = \|Q^* - Q^{\pi_k}\|_\infty.$$

Using part (a), derive an upper bound on e_k in terms of e_0, γ, and k. Explain what happens to e_k as $k \to \infty$.

Hint: If two functions satisfy $0 \leq f(s,a) \leq g(s,a)$ for every state-action pair (s,a), then $\|f\|_\infty \leq \|g\|_\infty$. Use this fact in part (a), step iii.